

Table 5- 15

| Bit | Value | Description            |
|-----|-------|------------------------|
| 8   | 0     | Motor continues to run |
|     | 1     | pause (media player)   |

#### status word

|             |    |       |             |                |             |   |     |
|-------------|----|-------|-------------|----------------|-------------|---|-----|
| 15          | 14 | 13    | 12          | 11             | 10          | 9 | 0   |
| (see 3.1.3) | -  | Speed | (see 3.1.3) | Target reached | (see 3.1.3) |   |     |
| MSB         |    |       |             |                |             |   | LSB |

In profile speed mode, the following bits in object 6041<sub>h</sub> (0381<sub>h</sub>) have a special function:

Table 5- 16

| Bit | Value | Description                                                                                                                                                                                 |
|-----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10  | 0     | Non-suspended state (bit8 of 6040 <sub>h</sub> (0380 <sub>h</sub> ) = 0): target speed not reached<br>Pause state (bit8 = 1 for 6040 <sub>h</sub> (0380 <sub>h</sub> )): motor deceleration |
|     | 1     | Non-suspended state (bit8 of 6040 <sub>h</sub> (0380 <sub>h</sub> ) = 0): target speed reached<br>Pause state (bit8 = 1 for 6040 <sub>h</sub> (0380 <sub>h</sub> )): motor speed 0          |
| 12  | 0     | Velocity is not zero.                                                                                                                                                                       |
|     | 1     | Speed is zero.                                                                                                                                                                              |

#### 5.5.4 Functional Description

In this mode, the upper controller sends the target speed, acceleration, and deceleration to the driver, and the speed and torque adjustments are performed internally.

#### 5.5.5 Configuration Example

- Configuration Mode  
Write 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) = 0, run mode 6060<sub>h</sub> (03C2<sub>h</sub>) = 0x03 to operate in contour speed mode;  
Send: 01 10 00 B1 00 01 02 00 00 (set to CIA402 mode)  
Send: 01 10 03 C2 00 01 02 00 03 (set to profile speed mode)
- Configuration parameters:  
Write target speed: 60FF<sub>h</sub> (0448<sub>h</sub>) = 50000;  
Send: 01 10 04 48 00 02 04 00 00 C3 50 (set target speed to 300 rpm)  
Write Profile Acceleration: 6083<sub>h</sub> (03FC<sub>h</sub>) = 40000.  
Send: 01 10 03 FC 00 02 04 00 00 9C 40 (sets acceleration 6083<sub>h</sub> (03FC<sub>h</sub>) to 40000)  
Write contour deceleration: 6084<sub>h</sub> (03FE<sub>h</sub>) = 40000.  
Send: 01 10 03 FE 00 02 04 00 00 9C 40 (set deceleration 6084<sub>h</sub> (03FE<sub>h</sub>) to 40000)
- Write the word control:  
6040<sub>h</sub> = 0x06→0x07→0x0F, motor running;  
Send: 01 10 03 80 00 01 02 00 06 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x6 to make motor ready)

Send: 01 10 03 80 00 01 02 00 07 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x7 to disable motor)

Send: 01 10 03 80 00 01 02 00 0F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0xF to enable motor)

- Parameter monitoring:

Current actual speed 606C<sub>h</sub> (03D5<sub>h</sub>) (unit: rpm)

## 5.6 Profile Torque Mode (PT)

In this mode, the host controller sends the target torque 6071<sub>h</sub> (03DB<sub>h</sub>), the torque ramp constant 6087<sub>h</sub> (0403<sub>h</sub>) to the drive, the torque adjustment is performed internally, and when the speed reaches the limiting value it will enter the speed control phase.

### 5.6.1 Structure Plan

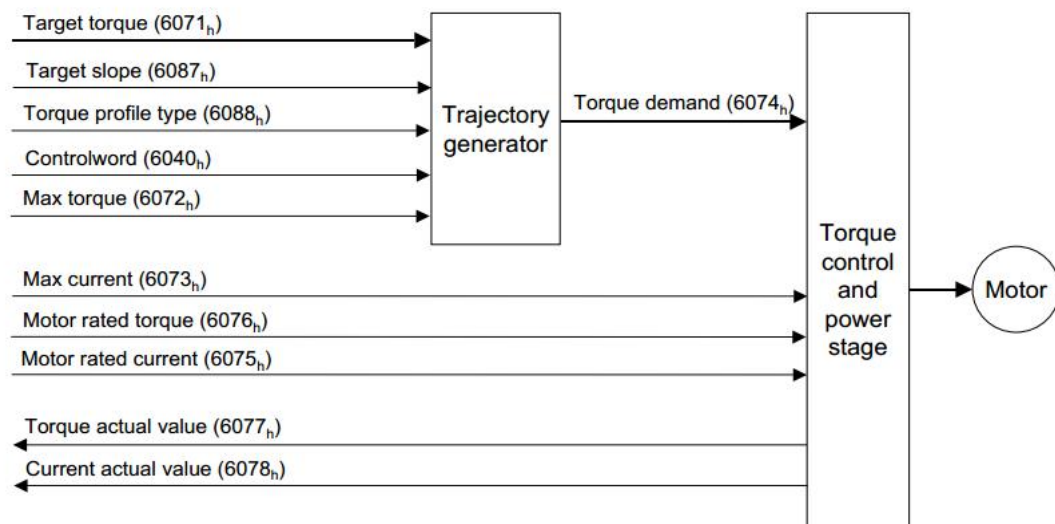


Figure 5- 17 Control block diagram

### 5.6.2 Relevant Object

The following objects require attention in this mode:

Table 5- 17

| Object Index                                            | Description                                                                                    |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 605D <sub>h</sub> (03C0) <sub>h</sub>                   | Pause mode selection                                                                           |
| 605A <sub>h</sub> (03BF) <sub>h</sub>                   | Quick stop mode selection                                                                      |
| 6071 <sub>h</sub> (03DB) <sub>h</sub>                   | Setting target torque (unit: 0.1%)                                                             |
| 6072 <sub>h</sub> (03DC) <sub>h</sub>                   | Maximum torque on the entire ramp (acceleration, conservation of torque, braking) (unit: 0.1%) |
| 2007 <sub>h</sub> : 10 <sub>h</sub> (016D) <sub>h</sub> | Torque control forward speed limit value (rpm)                                                 |
| 2007 <sub>h</sub> : 11 <sub>h</sub> (016E) <sub>h</sub> | Torque control reverse speed limit value (rpm)                                                 |
| 6087 <sub>h</sub> (0403) <sub>h</sub>                   | Maximum change in torque per second (unit: 0.1%)                                               |
| 6088 <sub>h</sub> (0405) <sub>h</sub>                   | Torque ramp type (0-slope, 2-none)                                                             |
| 6077 <sub>h</sub> (03E3) <sub>h</sub>                   | Actual value of torque (unit: 0.1%)                                                            |

### 5.6.3 Control Commands and Status Messages

#### start

To enable this mode, the value "0" must be set in object 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub> ) and "4" in object 6060<sub>h</sub> (03C2<sub>h</sub> ).

#### control word

|             |   |      |             |          |   |   |             |
|-------------|---|------|-------------|----------|---|---|-------------|
| 15          | 9 | 8    | 7           | 6        | 4 | 3 | 0           |
| (see 3.1.2) |   | Halt | (see 3.1.2) | reserved |   |   | (see 3.1.2) |
| MSB         |   |      |             | LSB      |   |   |             |

In profile torque mode, the following bits in object 6040<sub>h</sub> (0380<sub>h</sub> ) have a special function

Table 5- 18

| Bit | Value | Description            |
|-----|-------|------------------------|
| 8   | 0     | Motor continues to run |
|     | 1     | pause (media player)   |

#### status word

|             |    |          |    |             |                |   |             |
|-------------|----|----------|----|-------------|----------------|---|-------------|
| 15          | 14 | 13       | 12 | 11          | 10             | 9 | 0           |
| (see 3.1.3) |    | reserved |    | (see 3.1.3) | Target reached |   | (see 3.1.3) |
| MSB         |    |          |    | LSB         |                |   |             |

In profile torque mode, the following bits in object 6041<sub>h</sub> (0381<sub>h</sub> ) have a special function:

Table 5- 19

| Bit | Value | Description                                                                                                                                                                                          |
|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10  | 0     | Non-suspended state (bit8 = 0 for 6040 <sub>h</sub> (0380 <sub>h</sub> )): target torque not reached<br>In the pause state (bit8 = 1 for 6040 <sub>h</sub> (0380 <sub>h</sub> )): Motor deceleration |
|     | 1     | Non-suspended state (bit8 = 0 for 6040 <sub>h</sub> (0380 <sub>h</sub> )): target torque reached<br>Pause state (bit8 = 1 for 6040 <sub>h</sub> (0380 <sub>h</sub> )): motor speed 0                 |

### 5.6.4 Functional Description

#### Speed Limit Function

Running an integrated low-voltage servo motor in contour torque mode automatically switches to speed mode control and controls the current speed to be within the speed limit when the motor's operating speed exceeds the value set by the speed limit (forward speed limit value 2007<sub>h</sub> : 10<sub>h</sub> (016D<sub>h</sub> ) and reverse speed limit value 2007<sub>h</sub> : 11<sub>h</sub> (016E<sub>h</sub> )). When the given target torque is detected to be less than the average torque at the current speed, the motor exits speed control and reverts to torque mode control.

### 5.6.5 Configuration Example

- Configuration Mode  
Write 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) = 0, run mode 6060<sub>h</sub> (03C2<sub>h</sub>) = 0x04 to operate in profile torque mode;  
Send: 01 10 00 B1 00 01 02 00 00 (set to CIA402 mode)  
Send: 01 10 03 C2 00 01 02 00 04 (set to profile torque mode)
- Configuration parameters:  
Write target torque: 6071<sub>h</sub> (03DB<sub>h</sub>) = 500 (unit: 0.1%);  
Send: 01 10 03 DB 00 01 02 01 F4 (set target torque 6071<sub>h</sub> (03DB<sub>h</sub>) to 50%)  
Write Torque Ramp Type 6088<sub>h</sub> (0405<sub>h</sub>)=2;  
Write torque ramp 6087<sub>h</sub> (0403<sub>h</sub>)=10.  
Send: 01 10 04 05 00 01 02 00 02 (sets ramp type 6088<sub>h</sub> (0405<sub>h</sub>) to no ramp)  
Send: 01 10 04 03 00 02 04 00 00 0A (set torque ramp 6087<sub>h</sub> (0403<sub>h</sub>) to 10)
- Write the word control:  
6040<sub>h</sub> (0380<sub>h</sub>) = 0x06→0x07→0x0F, motor running;  
Send: 01 10 03 80 00 01 02 00 06 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x6 to make motor ready)  
Send: 01 10 03 80 00 01 02 00 07 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x7 to disable motor)  
Send: 01 10 03 80 00 01 02 00 0F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0xF to enable motor)
- Parameter monitoring:  
Actual torque 6077<sub>h</sub> (03E3<sub>h</sub>) (unit 0.1%)

## 5.7 Homing Mode (HM)

Homing mode is used to move from the current position to the home position of the device. During the movement, maximum acceleration, minimum deceleration, maximum velocity, minimum velocity, etc. are taken into account.

### 5.7.1 Structure Plan

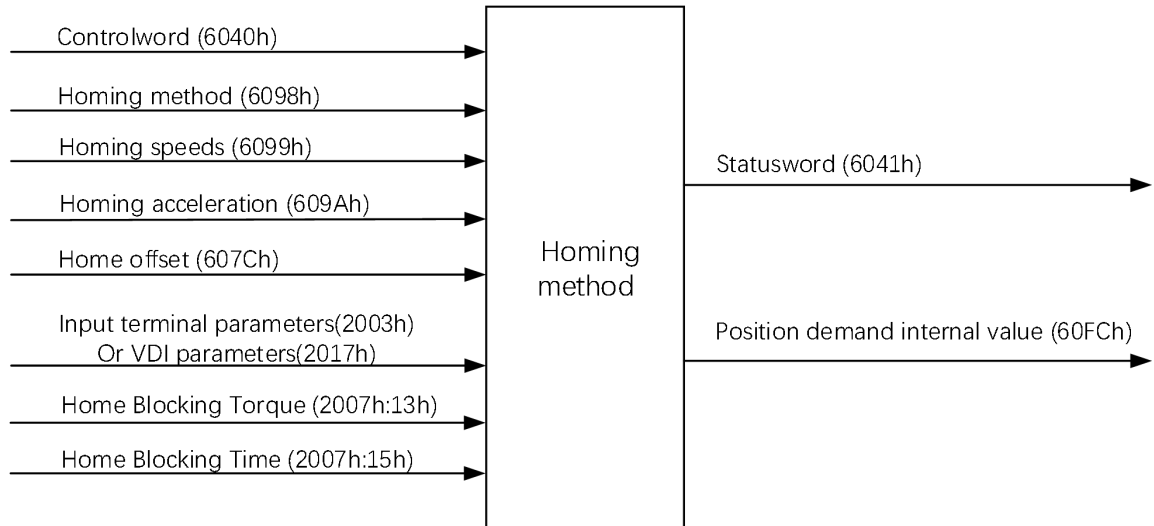


Figure 5- 18

### 5.7.2 Relevant Object

The following objects require attention in this mode:

Table 5- 20

| Object Index                                                                                                     | Description                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 607C <sub>h</sub> (03ED) <sub>h</sub>                                                                            | Specific description: Set the physical position of the mechanical zero point deviation from the motor home position in the home return mode; takes effect when bit15=1 of status word 6041 <sub>h</sub> (0381 <sub>h</sub> ) at the completion of the home return operation |
| 6098 <sub>h</sub> (0416) <sub>h</sub>                                                                            | Home return method (see "6.7.4 Functional description")                                                                                                                                                                                                                     |
| 6099 <sub>h</sub> :01 <sub>h</sub> (0417) <sub>h</sub><br>6099 <sub>h</sub> :02 <sub>h</sub> (0419) <sub>h</sub> | Running speed (user unit/s)<br>Sub-index: 6099 <sub>h</sub> :01 <sub>h</sub> (0417 <sub>h</sub> ): speed to find the switch;<br>6099 <sub>h</sub> :02 <sub>h</sub> (0419 <sub>h</sub> ): speed to find the origin                                                           |
| 609A <sub>h</sub> (041B) <sub>h</sub>                                                                            | Acceleration of the origin return run (user units/s <sup>2</sup> )                                                                                                                                                                                                          |
| 60FC <sub>h</sub> (0446) <sub>h</sub>                                                                            | Output of the trajectory generator, i.e. internally planned real-time position commands (encoder units)                                                                                                                                                                     |
| 6063 <sub>h</sub> (03C6) <sub>h</sub>                                                                            | Feedback of the current absolute position of the motor (encoder unit)                                                                                                                                                                                                       |
| 2003 <sub>h</sub> (00D5 <sub>h</sub> ~00E6) <sub>h</sub>                                                         | Functional configuration of physical input terminals (see "6.1.1 Digital Input" for details)<br>Note: You cannot have multiple input terminals configured as the same switch at the same time, or you                                                                       |

| Object Index                                              | Description                                                                       |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------|
|                                                           | will report a terminal setup fault.                                               |
| 2017 <sub>h</sub> (0326 <sub>h</sub> ~0345 <sub>h</sub> ) | Configuration of the virtual terminal input function (see: "6.1.1 Digital input") |
| 2007 <sub>h</sub> :13 <sub>h</sub> (0170 <sub>h</sub> )   | Detecting torque while searching for home position in blocked rotation            |
| 2007 <sub>h</sub> :15 <sub>h</sub> (0172 <sub>h</sub> )   | Detection time when blocking rotation to find the home position                   |

### Examples of the meaning of speed

The following graph shows the change in velocity when using the origin regression method 18:

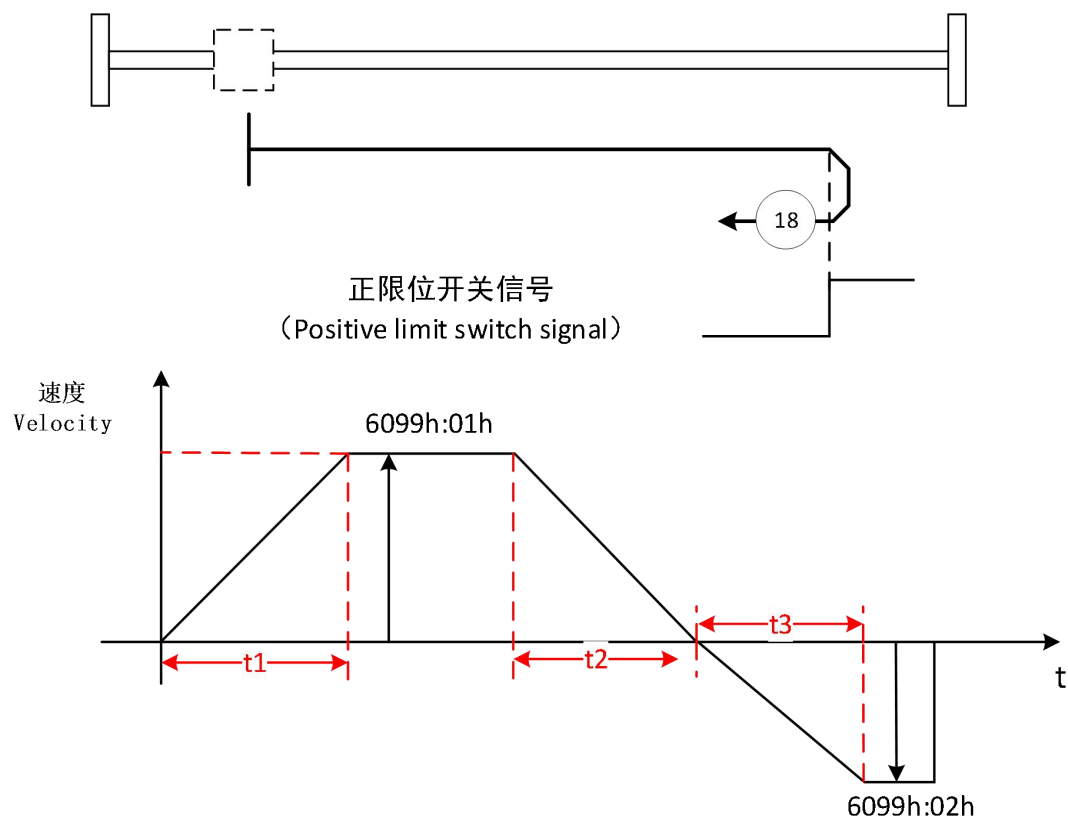


Figure 5- 19

$$\text{Among them: } t1 = \frac{6099h:01h(0417h)}{609Ah(041Bh)} ; t2 = \frac{6099h:01h(0417h)}{609Ah(041Bh)} ; t3 = \frac{6099h:02h(0419h)}{609Ah(041Bh)}$$

### 5.7.3 Control Commands and Status Messages

#### start

To enable this mode, the value "0" must be set in object 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) and "6" in object 6060<sub>h</sub> (03C2<sub>h</sub>), and the multifunction terminal must be configured with the function of home switch or limit switch. For details of the multifunction terminal configuration, see section 7.1.

#### control word

|             |      |             |              |                        |             |   |   |   |
|-------------|------|-------------|--------------|------------------------|-------------|---|---|---|
| 15          | 9    | 8           | 7            | 6                      | 5           | 4 | 3 | 0 |
| (see 3.1.2) | Halt | (see 3.1.2) | reserved (0) | Homing operation start | (see 3.1.2) |   |   |   |
| MSB         |      |             |              |                        | LSB         |   |   |   |

In home return mode, the following bits in object 6040<sub>h</sub> (0380<sub>h</sub>) have a special function

Table 5- 21

| Bit | Value | Description                                                                      |
|-----|-------|----------------------------------------------------------------------------------|
| 4   | 0     | Home return mode not activated                                                   |
|     | 1     | Initiation or continuation of home return mode                                   |
| 8   | 0     | The motor decides whether to start the home return or not by setting bit4.       |
|     | 1     | The motor is suspended at the setting of 605D <sub>h</sub> (03C0 <sub>h</sub> ). |

#### status word

|             |              |                 |             |                |             |   |   |
|-------------|--------------|-----------------|-------------|----------------|-------------|---|---|
| 15          | 14           | 13              | 12          | 11             | 10          | 9 | 0 |
| (see 3.1.3) | Homing error | Homing attained | (see 3.1.3) | Target reached | (see 3.1.3) |   |   |
| MSB         |              |                 |             |                | LSB         |   |   |

The following bits in object 6041<sub>h</sub> (0381<sub>h</sub>) have a special function in the home return mode:

Table 5- 22

| Bit | Value | Description                 |
|-----|-------|-----------------------------|
| 10  | 0     | Target position not reached |
|     | 1     | Target position reached     |
| 12  | 0     | Return of origin incomplete |
|     | 1     | Return of origin complete   |

## 5.7.4 Functional Description

### 5.7.4.1 Finding the home switch mode

The home return method using a switch currently supports 17~30 in CiA402. For home return to zero, we often need to start running from zero, and generally speaking, we need to run to the zero position command after home return to zero. For the convenience of users, this series of integrated motors will automatically return to the zero position after the home return.

**Configuration of the input terminal DI function number associated with the HM mode:**

Table 5- 23

| Function Number | Functional Definition   |
|-----------------|-------------------------|
| 14              | Positive Limit Switches |
| 15              | Negative Limit Switches |
| 31              | Home switch             |

**The directions and switch states that require origin return are shown in the table below:**

Table 5- 24

| Homing Method | Switch Related                     |
|---------------|------------------------------------|
| 17            | Negative Limit Switches            |
| 18            | Positive Limit Switches            |
| 19            | Home switch                        |
| 20            | Home switch                        |
| 21            | Home switch                        |
| 22            | Home switch                        |
| 23            | Home Switch, Positive Limit Switch |
| 24            | Home Switch, Positive Limit Switch |
| 25            | Home Switch, Positive Limit Switch |
| 26            | Home Switch, Positive Limit Switch |
| 27            | Home Switch, Negative Limit Switch |
| 28            | Home Switch, Negative Limit Switch |
| 29            | Home Switch, Negative Limit Switch |
| 30            | Home Switch, Negative Limit Switch |

- Homing Method 17:

Mechanical Home: Negative Limit Switch

When the negative limit switch is invalid, start back to zero at high speed in reverse, decelerate, reverse, and run at low speed in forward direction when the negative limit switch is valid (along the change), and stop when the negative limit switch is invalid (along the change).

When the negative limit switch is valid, the direct positive low speed starts back to zero and stops when it encounters the negative limit switch as invalid (along the change).

The motion trajectory is shown below:

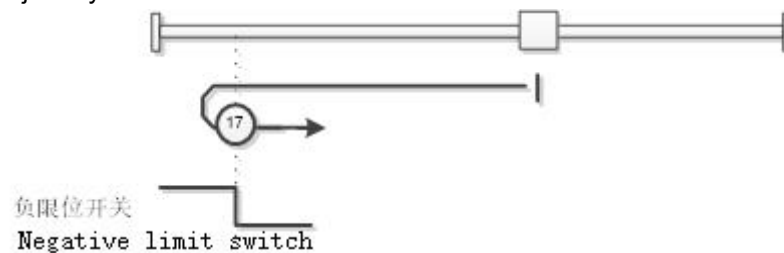


Figure 5- 20

- Homing Method 18:

Mechanical Home: Positive Limit Switch

When the positive limit switch is invalid, start back to zero at high speed in the positive direction, decelerate, reverse, reverse low speed when the positive limit switch is valid (along the change), and stop when the positive limit switch is invalid (along the change).

When the positive limit switch is valid, it starts back to zero at low speed in direct reverse, and stops when it encounters a positive limit switch that is invalid (along the change).

The motion trajectory is shown below:

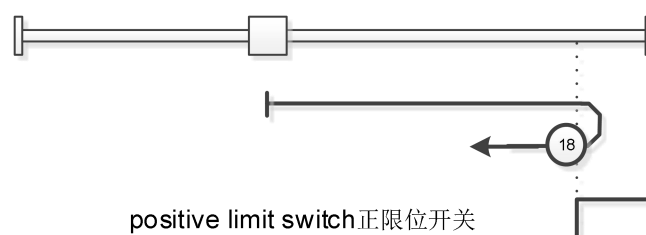


Figure 5- 21

- Homing Method 19 and 20:

Mechanical Home: Home Switch

Homing Method 19:

When the home switch is invalid, start back to zero with forward high speed, encounter the home switch is valid (along the change), decelerate, reverse, reverse low speed, encounter the home switch is invalid (along the change) to stop; when the home switch is valid, start back to zero with direct reverse low speed, encounter the home switch is invalid (along the change) to stop.

### Homing Method 20:

When the home switch is invalid, start back to zero at low speed directly in the forward direction, and stop when the home switch is valid (along the change); when the home switch is valid, start back to zero at high speed in the reverse direction, and stop when the home switch is invalid (along the change), decelerate, reverse, and run at low speed in the forward direction, and stop when the home switch is valid (along the change).

The motion trajectory is shown below:

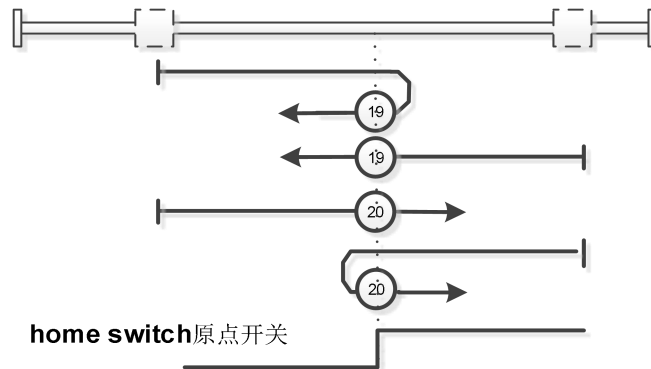


Figure 5- 22

### ● Homing Method 21 and 22:

#### Mechanical Home: Home Switch

### Homing Method 21:

When the home switch is invalid, start back to zero at high speed in reverse, decelerate, reverse, and run at low speed in forward direction when the home switch is invalid (along the change), and stop when the home switch is invalid (along the change); When the home switch is valid, start back to zero at low speed in forward direction directly, and stop when the home switch is invalid (along the change).

### Homing Method 22:

When the home switch is invalid, start back to zero at low speed in direct reverse, and stop when the home switch is valid (along the change); when the home switch is valid, start back to zero at high speed in forward direction, and stop when the home switch is invalid (along the change), decelerate, reverse, and run at low speed in reverse direction, and stop when the home switch is valid (along the change).

The motion trajectory is shown below:

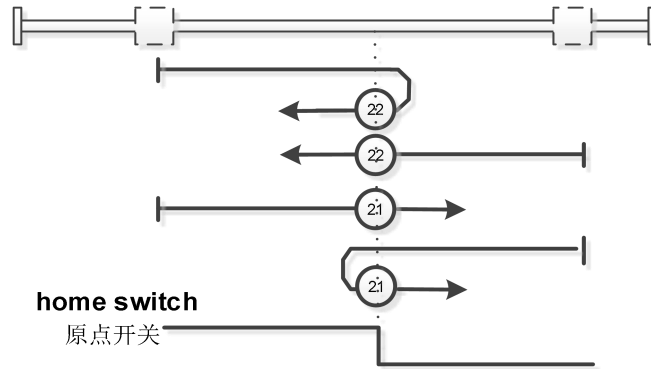


Figure 5- 23

● Homing Method 23-26:

Mechanical Home: Home Switch

These are actually motors moving in one direction first to scan the home switch. The only time the home switch is active is when it is a shorter search trajectory.

Homing Method 23:

- When the home switch is invalid, start back to zero at high speed in the positive direction without encountering the positive limit switch, decelerate, reverse, reverse low speed when the home switch is valid (along the change), and stop when the home switch is invalid (along the change);
- When the home switch is invalid, start back to zero with positive high speed, encounter positive limit switch, reverse, reverse high speed, encounter home switch valid (along the change), decelerate, continue to reverse low speed, encounter home switch invalid (along the change) stop;
- When the home switch is active, it starts back to zero at low speed in direct reverse and stops when it encounters the home switch is not active (along the change).

Homing Method 24 :

- When the home switch is invalid, start back to zero at high speed in the forward direction without encountering a positive limit switch, when the home switch is valid (along the change), decelerate, reverse, run at low speed in the reverse direction, when the home switch is invalid (along the change), reverse, run at low speed in the forward direction, and stop when the home switch is valid (along the change);
- When the home switch is invalid, start back to zero with positive high speed, encounter positive limit switch, reverse, reverse high speed, encounter home switch valid (along the change), decelerate, reverse low speed, encounter home switch invalid (along the change), reverse, positive low speed, encounter home switch valid (along the change) and stop;
- When the home switch is valid, start back to zero at low speed directly in reverse, reverse when the home switch is invalid (along the change), run at low speed in forward direction, and stop when the home switch is valid (along the change).

Homing Method 25:

- When the home switch is invalid, start back to zero at high speed in the forward direction

without encountering a positive limit switch, decelerate and run at low speed in the forward direction when encountering a home switch that is valid (along the change), reverse and run at low speed in the reverse direction when encountering a home switch that is invalid (along the change), and stop when encountering a home switch that is valid (along the change);

- b. When the home switch is invalid, start back to zero with positive high speed, encounter positive limit switch, reverse, reverse high speed, encounter home switch valid (along the change), decelerate, positive low speed, encounter home switch invalid (along the change), reverse, reverse low speed, encounter home switch valid (along the change) stop;
- c. When the home switch is valid, direct forward low speed start back to zero, encounter the home switch is invalid (along the change), reverse, reverse low speed, encounter the home switch is valid (along the change) stop.

Homing Method 26:

- a. When the home switch is invalid, start back to zero at a positive high speed without encountering a positive limit switch, decelerate and run at a positive low speed when the home switch is valid (along the change), and stop when the home switch is invalid (along the change);
- b. When the home switch is invalid, start back to zero with positive high speed, encounter positive limit switch, reverse, reverse high speed, encounter home switch valid (along the change), decelerate, positive low speed, encounter home switch invalid (along the change) stop;
- c. When the home switch is active, direct forward low speed starts back to zero and stops when it encounters a home switch that is not active (along the change).

The motion trajectory is shown below:

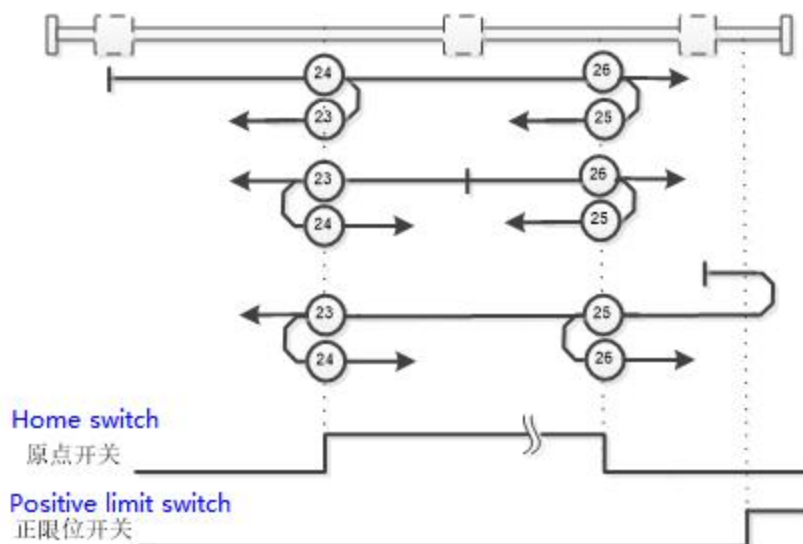


Figure 5- 24

#### ● Homing Method 27-30:

##### Mechanical Home: Home Switch

These are actually motors moving in one direction first to scan the home switch. The only time the home switch is active is when it is a shorter search trajectory.

## Homing Method 27:

- a. When the home switch is invalid, start back to zero at high speed in reverse without encountering a negative limit switch, decelerate, reverse, and run at low speed in forward direction when the home switch is valid (along the change), and stop when the home switch is invalid (along the change);
- b. When the home switch is invalid, start back to zero at high speed in reverse, encounter the negative limit switch, reverse, run at high speed in the positive direction, encounter the home switch valid (along the change), decelerate, continue to run at low speed in the positive direction, and stop when encountering the home switch invalid (along the change);
- c. When the home switch is active, direct forward low speed starts back to zero and stops when it encounters a home switch that is not active (along the change).

## Homing Method 28:

- a. When the home switch is invalid, start back to zero at high speed in reverse without encountering a negative limit switch, when the home switch is valid (along the change), decelerate, reverse, and run at low speed in the positive direction, when the home switch is invalid (along the change), reverse, and run at low speed in the reverse direction, and stop when the home switch is valid (along the change);
- b. When the home switch is invalid, start back to zero with reverse high speed, encounter negative limit switch, reverse, positive high speed operation, encounter home switch valid (along the change), decelerate, positive low speed operation, encounter home switch invalid (along the change), reverse, reverse low speed operation, encounter home switch valid (along the change) stop;
- c. When the home switch is valid, direct forward low speed start back to zero, encounter the home switch is invalid (along the change), reverse, reverse low speed, encounter the home switch is valid (along the change) stop.

## Homing Method 29:

- a. When the home switch is invalid, start back to zero at high speed in reverse without encountering a negative limit switch, when the home switch is valid (along the change), decelerate and run at low speed in reverse, when the home switch is invalid (along the change), reverse and run at low speed in positive direction, and stop when the home switch is valid (along the change);
- b. When the home switch is invalid, start back to zero with reverse high speed, encounter negative limit switch, reverse, positive high speed operation, encounter home switch valid (along the change), decelerate, reverse low speed operation, encounter home switch invalid (along the change), reverse, positive low speed operation, encounter home switch valid (along the change) stop;
- c. When the home switch is valid, start back to zero at low speed directly in reverse, reverse when the home switch is invalid (along the change), run at low speed in forward direction, and stop when the home switch is valid (along the change).

## Homing Method 30:

- a. When the home switch is invalid, start back to zero at high speed in reverse without encountering the negative limit switch, decelerate when the home switch is valid (along

the change), run at low speed in reverse, and stop when the home switch is invalid (along the change);

- b. When the home switch is invalid, start back to zero with reverse high speed, encounter negative limit switch, reverse, positive high speed operation, encounter home switch valid (along the change), decelerate, reverse low speed operation, encounter home switch invalid (along the change) stop;
- a. When the home switch is active, it starts back to zero at low speed in direct reverse and stops when it encounters the home switch is not active (along the change).

The motion trajectory is shown below:

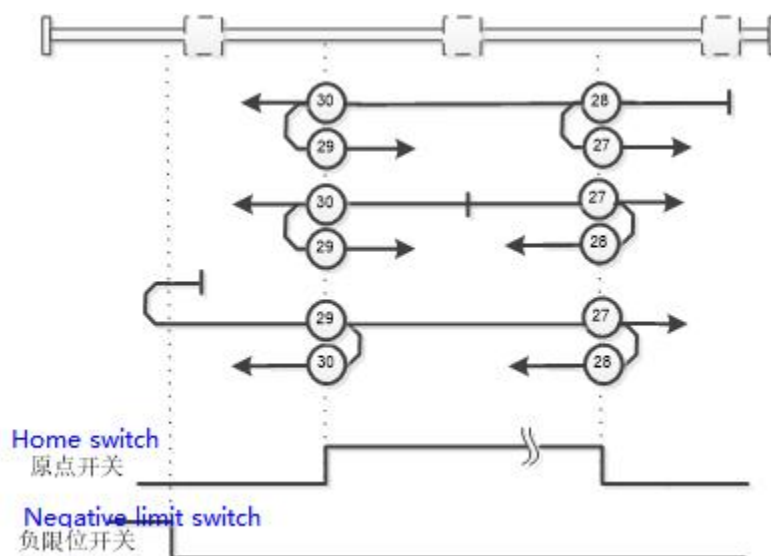


Figure 5- 25

#### 5.7.4.2 Blocking the turn to find the way of origin

The blocking home return method is currently only supported in closed-loop control mode. The blocking home return method is similar to the previous method of using a switch to find the home position, but the difference is that the blocking detection torque value is used instead of a limit switch.

The way of its origin return and the direction of its operation are shown in the table below:

Table 5- 25

| Homing Method | Running Direction                                                                |
|---------------|----------------------------------------------------------------------------------|
| 37            | Running at 6099 <sub>h</sub> :02 <sub>h</sub> (0419 <sub>h</sub> ) speed forward |
| 38            | Running at 6099 <sub>h</sub> :02 <sub>h</sub> (0419 <sub>h</sub> ) speed reverse |

In addition this method requires the following two parameters to be set:

- 1、Detection torque during home search in stalled rotation 2007<sub>h</sub>:13<sub>h</sub> (0170<sub>h</sub>)

This blocking torque threshold should be greater than the actual operating torque during the origin finding process and should generally be less than the maximum torque to avoid triggering the blocking fault protection.

- 2、Detection time when blocking to find the home position 2007<sub>h</sub>:15<sub>h</sub> (0172<sub>h</sub>)

This time is the time for the motor blocking to find the home position process to maintain the blocking.

The process of the specific plug-and-go home return method is schematically shown below:

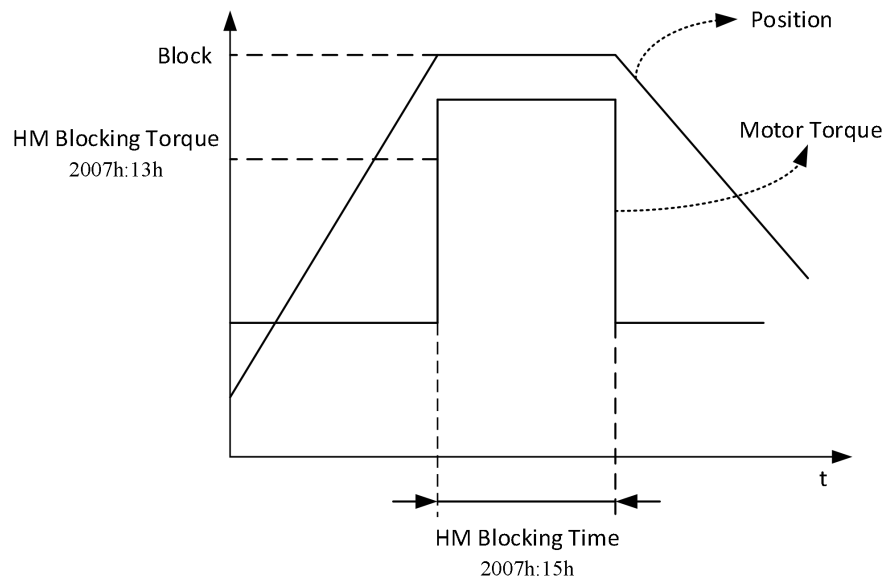


Figure 5- 26

### 5.7.5 Configuration Example

#### 1. Negative Limit Switch Finding Home Mode

- Setting Terminal  
 $2003_h : 03_h (00D5_h) = 15$  (negative limit switch),  $2003_h : 04_h (00D6_h) = 0$  (active low);  
 Send: 01 10 00 D5 00 01 02 00 0F (sets DI1 function as negative limit switch)  
 Send: 01 10 00 D6 00 01 02 00 00 (sets DI1 logic active low)
- Setup Mode  
 Write  $2002_h : 01_h (00B1_h) = 0$ , run mode  $6060_h (03C2_h) = 0x06$  to make it work in home return mode;  
 Send: 01 10 00 B1 00 01 02 00 00 (set to CIA402 mode)  
 Send: 01 10 03 C2 00 01 02 00 06 (set to home return mode)
- Setting the origin return method  
 $6098_h (0416_h) = 17$ .  
 Send: 01 10 04 16 00 01 02 00 11 (set home return mode to 17)
- Write find limit switch speed and find home signal speed  
 $6099_h : 01_h (0417_h) = 10000$   $6099_h : 02_h (0419_h) = 1000$ ;  
 Send: 01 10 04 17 00 02 04 00 00 27 10 (set seek limit switch speed to 10000)  
 Send: 01 10 04 19 00 02 04 00 00 03 E8 (set find home signal speed to 1000)
- Setting the return-to-zero acceleration  
 $609A_h (041B_h) = 200000$ ;  
 Send: 01 10 04 1B 00 02 04 00 03 0D 40 (set return to zero acceleration to 200000)
- Write control words  
 $6040_h (0380_h) = 0x06 \rightarrow 0x07 \rightarrow 0x0F \rightarrow 0x1F$ , motor running;

Send: 01 10 03 80 00 01 02 00 06 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x6 to make motor ready)

Send: 01 10 03 80 00 01 02 00 07 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x7 to disable motor)

Send: 01 10 03 80 00 01 02 00 0F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0xF to enable motor)

Send: 01 10 03 80 00 01 02 00 1F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x1F to run the motor)

## 2. Blocking the turn to find the way of origin

- Setting the origin return method

6098<sub>h</sub> (0416<sub>h</sub>) = 37.

Send: 01 10 04 16 00 01 02 00 25 (set home return mode to 37)

- Setting the blocking detection torque and blocking detection time

2007<sub>h</sub> : 13<sub>h</sub> (0170<sub>h</sub>) = 300, 2007<sub>h</sub> : 15<sub>h</sub> (0172<sub>h</sub>) = 50

Send: 01 10 01 70 00 01 02 01 2C (set blocking detection torque to 30%)

Send: 01 10 01 72 00 01 02 00 32 (set blocking detection time to 50ms)

- Setup Mode

Write 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) = 0, run mode 6060<sub>h</sub> (03C2<sub>h</sub>) = 0x06 to make it work in home return mode;

Send: 01 10 00 B1 00 01 02 00 00 (set to CIA402 mode)

Send: 01 10 03 C2 00 01 02 00 06 (set to home return mode)

- Setting the Find Home Signal Speed

6099<sub>h</sub> : 02<sub>h</sub> = 1000.

Send: 01 10 04 19 00 02 04 00 00 03 E8 (set find home signal speed to 1000)

- Setting the return-to-zero acceleration

609A<sub>h</sub> (041B<sub>h</sub>) = 200000;

Send: 01 10 04 1B 00 02 04 00 03 0D 40 (set return to zero acceleration to 200000)

- Write control words

6040<sub>h</sub> = 0x06→0x07→0x0F→0x1F, motor running

Send: 01 10 03 80 00 01 02 00 06 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x6 to make motor ready)

Send: 01 10 03 80 00 01 02 00 07 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x7 to disable motor)

Send: 01 10 03 80 00 01 02 00 0F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0xF to enable motor)

Send: 01 10 03 80 00 01 02 00 1F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x1F to run the motor)

## 5.8 Interpolation mode (IP)

The interpolated position mode is used to synchronize multiple axes. To do this, the advanced controller performs slope and path calculations and passes to the controller the corresponding demand positions that the axes are at a particular time. The controller interpolates between these intermediate position points.

### 5.8.1 Structure Plan

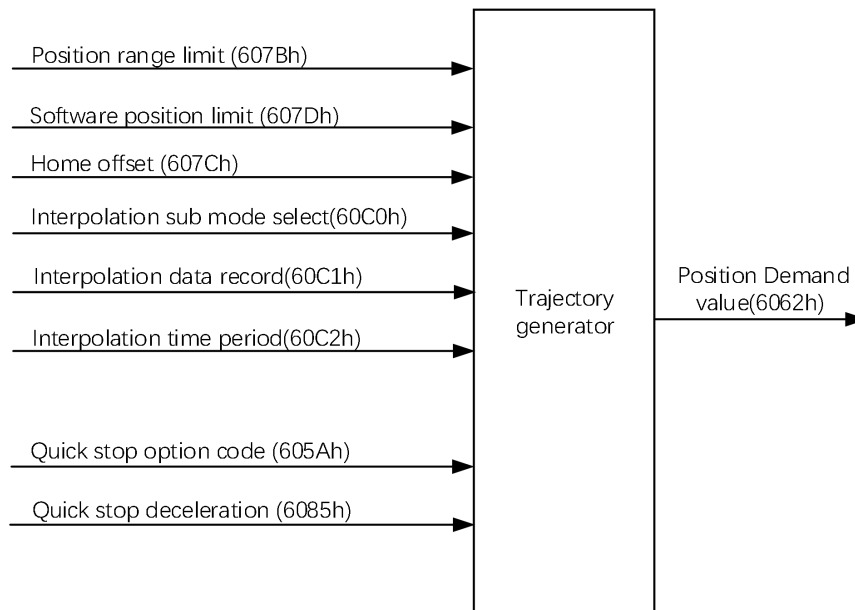


Figure 5- 27

### 5.8.2 Relevant Object

The following objects require attention in this mode:

Table 5- 26

| Object Index                                                                                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 607D <sub>h</sub> : 01 <sub>h</sub> (03EF ) <sub>h</sub><br>607D <sub>h</sub> : 02 <sub>h</sub> (03F1 ) <sub>h</sub> | Limit value for the target position (user units).                                                                                                                                                                                                                                                                                                                                                                            |
| 60C0 <sub>h</sub> (042C ) <sub>h</sub>                                                                               | Interpolation type selection, this device only supports linear interpolation, the default value is 0                                                                                                                                                                                                                                                                                                                         |
| 60C1 <sub>h</sub> : 01 <sub>h</sub> (042D ) <sub>h</sub>                                                             | Setting the interpolation target position (user units)                                                                                                                                                                                                                                                                                                                                                                       |
| 60C2 <sub>h</sub> : 01 <sub>h</sub> (042F ) <sub>h</sub><br>60C2 <sub>h</sub> : 02 <sub>h</sub> (0430 ) <sub>h</sub> | Interpolation time (1-20ms recommended)<br>Defined jointly by 60C2 <sub>h</sub> : 01 <sub>h</sub> (042F <sub>h</sub> ) and 60C2 <sub>h</sub> : 02 <sub>h</sub> (0430 <sub>h</sub> ),<br>interpolation period = t*10 <sup>n</sup> seconds<br>60C2 <sub>h</sub> : 01 <sub>h</sub> (042F <sub>h</sub> ): interpolation time constant t.<br>60C2 <sub>h</sub> : 02 <sub>h</sub> (0430 <sub>h</sub> ): interpolation time index n |
| 605A <sub>h</sub> (03BF ) <sub>h</sub>                                                                               | Quick stop mode selection                                                                                                                                                                                                                                                                                                                                                                                                    |
| 605D <sub>h</sub> (03C0 ) <sub>h</sub>                                                                               | Pause mode selection                                                                                                                                                                                                                                                                                                                                                                                                         |
| 6085 <sub>h</sub> (0400 ) <sub>h</sub>                                                                               | Stopping deceleration rate when performing a "quick stop" (user unit/s <sup>2</sup> )                                                                                                                                                                                                                                                                                                                                        |
| 6062 <sub>h</sub> (03C4 ) <sub>h</sub>                                                                               | Target position command value currently in effect within the drive (user units)                                                                                                                                                                                                                                                                                                                                              |

| Object Index                          | Description                                                           |
|---------------------------------------|-----------------------------------------------------------------------|
| 6063 <sub>h</sub> (03C6) <sub>h</sub> | Feedback of the current absolute position of the motor (encoder unit) |

### 5.8.3 Control Commands and Status Messages

#### start

To enable this mode, you must set the value "0" in object 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) and "7" in object 6060<sub>h</sub> (03C2<sub>h</sub>).

#### control word

|             |      |             |          |                      |             |
|-------------|------|-------------|----------|----------------------|-------------|
| 15          | 9    | 8 7         | 6 5      | 4                    | 3 0         |
| (see 3.1.2) | Halt | (see 3.1.2) | reserved | Enable interpolation | (see 3.1.2) |
| MSB         |      |             |          |                      | LSB         |

In interpolation mode, the following bits in object 6040<sub>h</sub> (0380<sub>h</sub>) have a special function:

Table 5- 27

| Bit | Value | Definition                                                                          |
|-----|-------|-------------------------------------------------------------------------------------|
| 4   | 0     | disablement interpolation                                                           |
|     | 1     | Enable interpolation                                                                |
| 8   | 0     | Execute the instruction in bit 4                                                    |
|     | 1     | Configuration suspension according to object 605D <sub>h</sub> (03C0 <sub>h</sub> ) |

#### status word

|             |                 |                |             |                |             |   |     |
|-------------|-----------------|----------------|-------------|----------------|-------------|---|-----|
| 15          | 14              | 13             | 12          | 11             | 10          | 9 | 0   |
| (see 3.1.3) | following error | ip mode active | (see 3.1.3) | Target reached | (see 3.1.3) |   |     |
| MSB         |                 |                |             |                |             |   | LSB |

In interpolation mode, the following bits in object 6041<sub>h</sub> (0381<sub>h</sub>) have a special function:

Table 5- 28

| Bit | Value | Definition                         |
|-----|-------|------------------------------------|
| 10  | 0     | Target position not reached        |
|     | 1     | Reach target position              |
| 12  | 0     | The interpolation is not activated |
|     | 1     | interpolation activation           |
| 13  | 0     | No following error                 |
|     | 1     | With following error               |

### 5.8.4 Functional Description

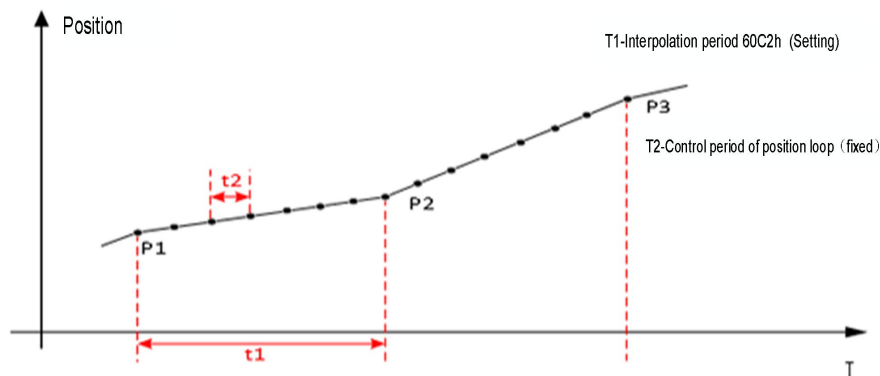


Figure 5- 28 Schematic diagram of interpolation

**Note:** t1: interpolation period, which can be set via the object dictionaries 60C2<sub>h</sub> : 01<sub>h</sub> (042F<sub>h</sub>) and 60C2<sub>h</sub> : 02<sub>h</sub> (0430<sub>h</sub>) (1-20ms recommended).

t2: position loop control period, determined internally by the drive.

P1/P2/P3: Absolute Position, Absolute Position instruction is sent via object dictionary 60C1<sub>h</sub> : 01<sub>h</sub> (042D<sub>h</sub>), Interpolation mode only supports Absolute Position instruction.

### 5.8.5 Configuration Example

- Configure to work in interpolation mode:  
Write 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) = 0, run mode 6060<sub>h</sub> (03C2<sub>h</sub>) = 0x07;  
Send: 01 10 00 B1 00 01 02 00 00 (set to CIA402 mode)  
Send: 01 10 03 C2 00 01 02 00 07 (set to interpolation mode)
- Configure the interpolation time:  
The time constant 60C2<sub>h</sub> : 01<sub>h</sub> (042F<sub>h</sub>) = 20 with the time exponent 60C2<sub>h</sub> : 02<sub>h</sub> (0430<sub>h</sub>) = -3 and the time is 20ms;  
Send: 01 10 04 2F 00 01 02 00 14 (set time constant to 20)  
Send: 01 10 04 30 00 01 02 00 FD (set time index to -3)
- The motor runs:  
Write control word 6040<sub>h</sub> (0380<sub>h</sub>) = 0x06→0x07→0x0F→0x1F  
Send: 01 10 03 80 00 01 02 00 06 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x6 to make motor ready)  
Send: 01 10 03 80 00 01 02 00 07 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x7 to disable motor)  
Send: 01 10 03 80 00 01 02 00 0F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0xF to enable motor)  
Send: 01 10 03 80 00 01 02 00 1F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x1F to run the motor)
- The host computer writes the interpolation position according to the interpolation cycle 60C1<sub>h</sub> : 01<sub>h</sub> (042D<sub>h</sub>) (Only absolute position instruction is supported.)